ECON2050 Tutorial 9

Optimization II: Linear Programming

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Today's Plan

- Recap: key ideas from Lecture 9
- Tutorial questions: Q1–Q4

Tutorial Q1

Question 1

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be given by

$$f(x, y) = x - y.$$

Solve the following problems graphically:

(a) $\max_{x,y \in \mathbb{R}} f(x, y)$;

(b) $\max_{x,y \in \mathbb{R}} f(x, y)$ subject to $x^2 + y^2 = 8$;

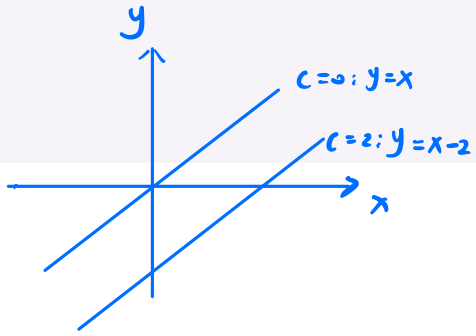
(c) $\max_{x \geq 0,\, y \geq 0} f(x, y)$ subject to $x^2 + y^2 \leq 8$.

For each output, there is a level curve with level $f(x,y)$,

$f(x,y) = x - y$

$c = x - y \Rightarrow y = x - c$

Goal: find highest $f(x,y)$ $\Rightarrow$ find the highest level c



(a) $\max_{x,y} f(x,y) \Leftrightarrow \max c$

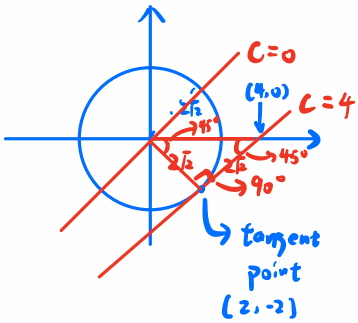
Without constraints, we can always shift the level curve downward.

$\Rightarrow$ c is unbounded

$\Rightarrow$ $f(x,y)$ can be infinitely large

$\Rightarrow$ $\max_{x,y} f(x,y)$ has no solution.

(b) $\max_{x,y} f(x,y)$ s.t $x^2+y^2=8$



Constraint $\{(x,y) \in \mathbb{R}^2 : x^2+y^2=8\}$

Circle centered at $(0,0)$ with radius $\sqrt{8} = 2\sqrt{2}$

Goal: find the highest attainable level c

has to intersect with the constraint set

The highest attainable level curve is the tangent line to that circle.

Pythagorean theorem: $\sqrt{(2\sqrt{2})^2+(2\sqrt{2})^2} = \sqrt{16} = 4$

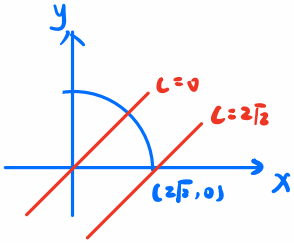
plug $(4,0)$ into $y = x - c$

$0 = 4 - c$

$c = 4$: optimal value

From the graph, the optimal point is $(2,-2)$

(c) $\max_{x,y} f(x,y)$ s.t $\{(x,y) \in \mathbb{R}^2_+ : x^2+y^2 \le 8\}$



The highese attainable curve intersects with the constraint set at $(2\sqrt{2}, 0)$

optimal point

plug $(2\sqrt{2},0)$ into $y = x - c$

$0 = 2\sqrt{2} - c$

$c = 2\sqrt{2}$: optimal value

Recap: Linear programming

Linear programming (LP)

A linear programming problem optimizes a *linear* objective function subject to linear inequality and equality constraints.

A standard maximization form is

$$\max_{\boldsymbol{x}} \ \boldsymbol{c} \cdot \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \qquad \boldsymbol{x} \geq \boldsymbol{0}.$$

The objective function is linear and the constraint set is determined by linear constraints.

Recap: Feasible set and convexity

Feasible point

A point is **feasible** if it belongs to the constraint set of the linear program.

For the standard form,

$$D = \{\boldsymbol{x} \in \mathbb{R}^n : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}\}.$$

Convexity of the constraint set

The constraint set of any linear program is convex because it is described by linear inequalities and equalities.

Tutorial Q1: Knowledge used

Knowledge used

- Level curves of the objective function.
- Graphical maximization over a constraint set.
- A maximization problem need not have a solution when the objective is unbounded above.
- For part (b), tangency between a level curve and the constraint boundary; the slope of the boundary can be obtained by the Implicit Function Theorem.

Tutorial Q1(a): Level curves and no maximum

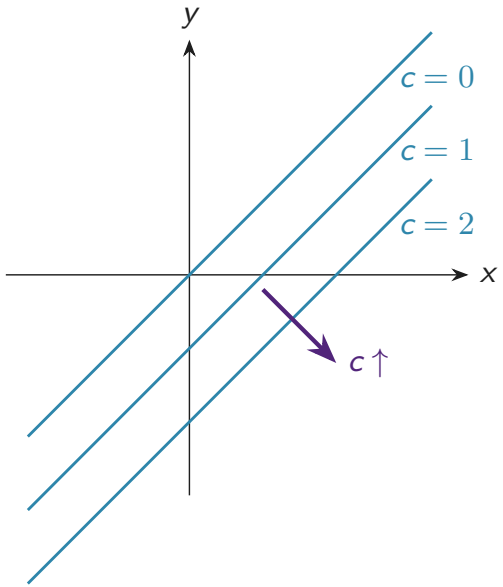
Solution

A level curve of level c satisfies

$$x - y = c \quad \Longleftrightarrow \quad y = x - c.$$

These are parallel lines with slope 1.
With no constraint on (x, y), we can always choose a larger level c. Hence the objective is unbounded above and

the maximization problem has no solution.



Tutorial Q1(b): Graphical solution

Solution

The constraint set is the circle

$$x^2 + y^2 = 8.$$

The highest attainable level curve of $f(x, y) = x - y$ is tangent to the circle at

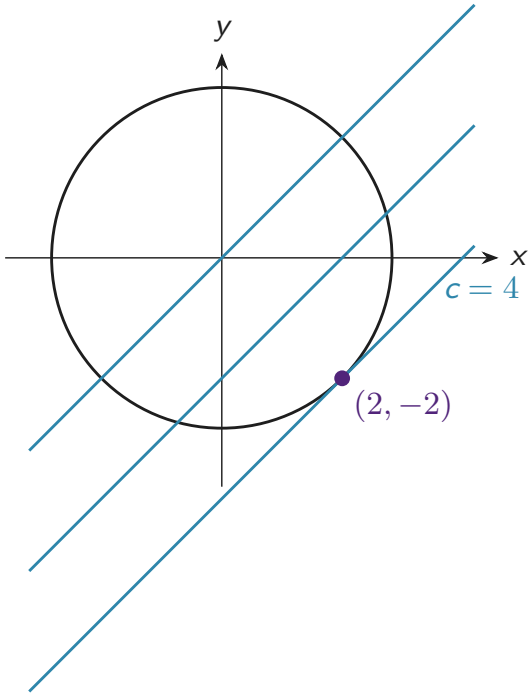
$$(2, -2).$$

At this point,

$$f(2, -2) = 4.$$

Hence

$(2, -2)$ is the maximizer, with maximum value 4.



Tutorial Q1(b): Tangency condition

Solution

By the Implicit Function Theorem, the slope of the tangent to

$$x^2 + y^2 = 8$$

at (x, y) is

$$\frac{dy}{dx} = -\frac{2x}{2y} = -\frac{x}{y}, \qquad y \neq 0.$$

The level curves $y = x - c$ have slope 1. At tangency,

$$-\frac{x}{y} = 1 \quad \Longleftrightarrow \quad x = -y.$$

Combining with $x^2 + y^2 = 8$ gives $x = \pm 2$. The maximum is attained at

$$(x, y) = (2, -2).$$

At $(-2, 2)$, f attains the minimum on the circle.

Tutorial Q1(c): First-quadrant constraint

Solution

Now the feasible set is

$$\{(x, y) : x \geq 0,\ y \geq 0,\ x^2 + y^2 \leq 8\}.$$

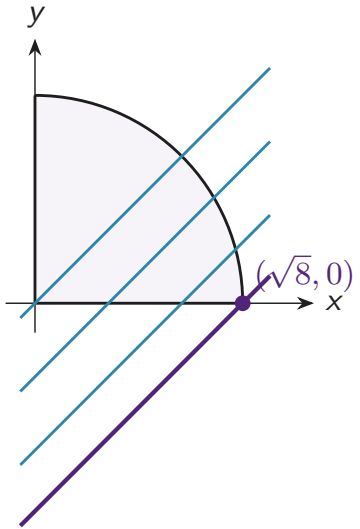
Moving the level curves $y = x - c$ toward larger c, the last feasible contact occurs at

$$(\sqrt{8}, 0).$$

Therefore,

$$(\sqrt{8}, 0) \text{ is the maximizer,}$$

with maximum value $\sqrt{8}$.



Tutorial Q2: Factory problem

Question 2 – setup

A factory produces orange juice and apple juice.

- Revenue per unit: 1 for orange juice and 2 for apple juice.
- Total production capacity: at most 2 thousand units.
- Labor: orange juice requires 1 unit; apple juice requires $\frac{8}{3}$ units; total labor available is 4 thousand units.
- The license limits orange-juice production to 1.5 thousand units.

Let x_1 and x_2 denote production of orange and apple juice, respectively, measured in thousands.

Tutorial Q2: Linear-program formulation

Question 2 – model

The factory's revenue-maximization problem is

$$\begin{aligned} \max \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + \frac{8}{3}x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1 \leq \frac{3}{2}, \\ & x_1, x_2 \geq 0. \end{aligned}$$

Tutorial Q2: Tasks

Question 2

For the linear program on the previous slide:

(a) Use the Extreme Value Theorem to determine whether the problem has a solution.

(b) Use the Fundamental Theorem of Linear Programming to find a solution.

(c) Write down the dual.

(d) Interpret the dual variables.

(e) Determine whether the optimal value of the dual is the same as the primal.

(a) ① $f(x_1, x_2) = x_1 + 2x_2$ is cts because the sum of two affine functions is cts.

② The constraint set is compact = closed + bounded

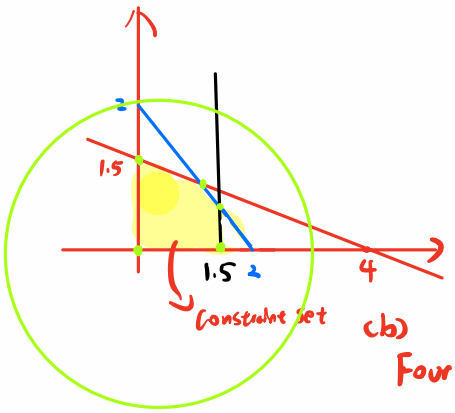
closed: it contained all of its boundary points

bounded: $\|(x,y)\| \leq 2$, $\forall (x,y) \in$ constraint

By EVT, the linear program problem has at least one solution.

$$\begin{aligned} \max \quad & x_1 + 2x_2 \\ \text{s.t.} \quad & x_1 + \frac{8}{3}x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1 \leq \frac{3}{2}, \end{aligned}$$

$$x_1, x_2 \geq 0.$$



Fundamental Theorem of Linear Programming

Suppose the linear program

$$\max\{\boldsymbol{c}\cdot\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \geq 0\}$$

has a solution. Then an extreme point of the constraint set is a solution.

Thus, once existence is known, we can search among the vertices of the feasible set.

(b)

Four vertices: $(0,0) \Rightarrow f(0,0) = 0$

$(0, 1.5) \Rightarrow f(0, 1.5) = 3$

$(1.5, 0) \Rightarrow f(1.5, 0) = 1.5$

$(0.8, 1.2) \Rightarrow f(0.8, 1.2) = 3.2$ Highest

$(1.5, 0.5) \Rightarrow f(1.5, 0.5) = 2.5$

$\Rightarrow$ The solution is $(0.8, 1.2)$ with the optimal value 3.2

$$\begin{cases} x_1 + \frac{8}{3}x_2 = 4 \\ x_1 + x_2 = 2 \end{cases} \Rightarrow \begin{cases} x_1 = 0.8 \\ x_2 = 1.2 \end{cases}$$

(c)

Consider the primal problem

$$\max\ \boldsymbol{c}\cdot\boldsymbol{x} \quad \text{s.t.} \quad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b},\quad \boldsymbol{x} \geq 0. \qquad \text{(P)}$$

Its dual is

$$\min\ \boldsymbol{y}\cdot\boldsymbol{b} \quad \text{s.t.} \quad \boldsymbol{y}\boldsymbol{A} \geq \boldsymbol{c},\quad \boldsymbol{y} \geq 0. \qquad \text{(D)}$$

The dual searches for the smallest upper bound on the primal objective value generated by non-negative linear combinations of the primal constraints.

$$\begin{aligned} \max\quad & x_1 + 2x_2 \\ \text{s.t.}\quad & x_1 + \frac{8}{3}x_2 \leq 4, \\ & x_1 + x_2 \leq 2, \\ & x_1 \leq \frac{3}{2}, \\ & x_1, x_2 \geq 0. \end{aligned}$$

$c = (1, 2)$ $\quad A = \begin{pmatrix} 1 & \frac{8}{3} \\ 1 & 1 \\ 1 & 0 \end{pmatrix}$ $\quad b = \begin{pmatrix} 4 \\ 2 \\ \frac{3}{2} \end{pmatrix}$

Primal problem: $\max\ c\cdot x$ s.t. $Ax \leq b,\ x \geq 0$

Dual problem: $\min\ y\cdot b$ s.t. $yA \geq c,\ y \geq 0$

$y = (y_1, y_2, y_3)$ is the dual variable vector

$$y\cdot b = (y_1, y_2, y_3)\cdot\begin{pmatrix} 4 \\ 2 \\ \frac{3}{2} \end{pmatrix} = 4y_1 + 2y_2 + \frac{3}{2}y_3$$

$$yA = (y_1, y_2, y_3)\cdot\begin{pmatrix} 1 & \frac{8}{3} \\ 1 & 1 \\ 1 & 0 \end{pmatrix} = (y_1 + y_2 + y_3,\ \frac{8}{3}y_1 + y_2 + 0) \geq (1, 2)$$

$y = (y_1, y_2, y_3) \geq (0, 0, 0)$

Dual: $\min\ 4y_1 + 2y_2 + \frac{3}{2}y_3$

s.t. $y_1 + y_2 + y_3 \geq 1$

$\frac{8}{3}y_1 + y_2 \geq 2$

$y_1, y_2, y_3 \geq 0$

(d) A dual variable measures the marginal value of the resource represented by that constraint

y_1: marginal value of labor

y_2: MV of the total production capacity

y_3: MV of the orange-juice licences limit

(e) Theorem: Strong duality

If either the primal or the dual has a solution, then the optimal value of the primal = the optimal value of the dual

By part (a) and (b), the primal has a solution with optimal value 3.2.

By strong duality, the optimal value of the dual is also 3.2.

Recap: Extreme Value Theorem

Existence of global extrema

If $D \subseteq \mathbb{R}^n$ is compact and $f : D \to \mathbb{R}$ is continuous, then f has at least one global maximum and at least one global minimum on D.

Convexity of the constraint set

The constraint set of any linear program is convex because it is described by linear inequalities and equalities.

Recap: Extreme points and Fundamental Theorem

Extreme point

Let $S \subseteq \mathbb{R}^n$ be convex. An extreme point of S is a point that cannot be expressed as a convex combination of other points in S.

For a linear program, the constraint set is a polyhedron, and its extreme points are its *corners* or *vertices*.

Fundamental Theorem of Linear Programming

Suppose the linear program

$$\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}\}$$

has a solution. Then an extreme point of the constraint set is a solution.

Thus, once existence is known, we can search among the vertices of the feasible set.

Recap: Primal and dual

Consider the primal problem

$$\max \ \boldsymbol{c} \cdot \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}, \quad \boldsymbol{x} \geq \boldsymbol{0}. \tag{P}$$

Its dual is

$$\min \ \boldsymbol{y} \cdot \boldsymbol{b} \qquad \text{s.t.} \qquad \boldsymbol{y}\boldsymbol{A} \geq \boldsymbol{c}, \quad \boldsymbol{y} \geq \boldsymbol{0}. \tag{D}$$

The dual searches for the smallest upper bound on the primal objective value generated by non-negative linear combinations of the primal constraints.

Shadow prices

At an optimum, a dual solution $\boldsymbol{y}^*$ can be interpreted as the *shadow prices* of the primal constraints.

For a sufficiently small relaxation of a primal constraint, the corresponding dual variable measures the per-unit change in the optimal objective value, holding the other constraints fixed.
Equivalently, a dual variable measures the marginal value of the resource represented by that constraint.

Recap: Weak and strong duality

Let V and W be the optimal values of the primal and dual, respectively.

Weak duality

For every primal-feasible $\boldsymbol{x}$ and dual-feasible $\boldsymbol{y}$,

$$\boldsymbol{c} \cdot \boldsymbol{x} \leq \boldsymbol{y} \cdot \boldsymbol{b}.$$

Hence,

$$V \leq W.$$

Strong duality

If either the primal or the dual has a solution, then

$$V = W.$$

Tutorial Q2: Knowledge used

Knowledge used

- Extreme Value Theorem: continuous objective + compact feasible set implies existence.
- Fundamental Theorem of Linear Programming: if a solution exists, a vertex is a solution.
- Standard primal-dual pair:

$$\max \ \boldsymbol{c} \cdot \boldsymbol{x} \quad \Longleftrightarrow \quad \min \ \boldsymbol{y} \cdot \boldsymbol{b}.$$

- Dual variables as marginal values / imputed resource costs.
- Strong duality: if a solution exists, the primal and dual optimal values coincide.

Tutorial Q2(a): Existence of a solution

Solution

The objective

$$f(x_1, x_2) = x_1 + 2x_2$$

is linear and therefore continuous.
The feasible set D is closed because all constraints are weak inequalities, and it is bounded by the production, labor, and license constraints. Hence D is compact.
By the Extreme Value Theorem,

the linear programming problem has a solution.

Tutorial Q2(b): Extreme points

Solution

By part (a), a solution exists. The Fundamental Theorem of Linear Programming therefore allows us to search among the extreme points of D.
The five extreme points are

$$(0,0), \qquad \left(0, \frac{3}{2}\right), \qquad \left(\frac{4}{5}, \frac{6}{5}\right),$$
$$\left(\frac{3}{2}, \frac{1}{2}\right), \qquad \left(\frac{3}{2}, 0\right).$$

We evaluate $f(x_1, x_2) = x_1 + 2x_2$ at these points.

Tutorial Q2(b): Compare objective values

Solution

$$
\begin{aligned}
f(0,0) &= 0, \\
f\left(0, \frac{3}{2}\right) &= 3, \\
f\left(\frac{4}{5}, \frac{6}{5}\right) &= \frac{16}{5}, \\
f\left(\frac{3}{2}, \frac{1}{2}\right) &= \frac{5}{2}, \\
f\left(\frac{3}{2}, 0\right) &= \frac{3}{2}.
\end{aligned}
$$

Since $\frac{16}{5}$ is the largest value,

$$
(x_1^*, x_2^*) = \left(\frac{4}{5}, \frac{6}{5}\right)
$$

is a solution, with optimal value $16/5$.

Tutorial Q2(c): Write down the dual

Solution

Let $y_1, y_2, y_3 \geq 0$ be the dual variables for the labor, total-capacity, and orange-juice constraints, respectively.
The dual is

$$\begin{aligned} \min \quad & 4y_1 + 2y_2 + \frac{3}{2}y_3 \\ \text{s.t.} \quad & y_1 + y_2 + y_3 \geq 1, \\ & \frac{8}{3}y_1 + y_2 \geq 2, \\ & y_1, y_2, y_3 \geq 0. \end{aligned}$$

Tutorial Q2(d)–(e): Interpretation and duality

Solution

(d) Interpretation.

- y_1: marginal value of labor;
- y_2: marginal value of total production capacity;
- y_3: marginal value of the orange-juice production limit.

Equivalently, they can be interpreted as the imputed costs of one additional unit of each resource.

(e) Optimal values. By part (a), the primal has a solution. Therefore, by strong duality,

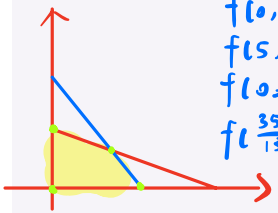
$$V = W.$$

Hence the optimal value of the dual is the same as the primal optimal value, $16/5$.

Tutorial Q3

Question 3

Consider



$f(0,0) = 0$
$f(5,0) = 10$
$f(0,3) = 21$ Highest
$f(\frac{35}{13}, \frac{24}{13}) = \frac{238}{13}$
$\Rightarrow$ 21 is the optimal value and (0,3) is the optimal point

$$\begin{aligned} \max \quad & 2x + 7y \\ \text{s.t.} \quad & 4x + 5y \le 20, \\ & 3x + 7y \le 21, \\ & x \ge 0,\ y \ge 0. \end{aligned}$$

20.1 $\uparrow$ $\Rightarrow C = (2,7)$

$\Rightarrow A = \begin{pmatrix} 4 & 5 \\ 3 & 7 \end{pmatrix}$, $b = \begin{pmatrix} 20 \\ 21 \end{pmatrix}$

Dual: min $z \cdot b$ s.t $z \cdot A \ge c$, $z \ge 0$
where $z = (z_1, z_2)$

20.8 $\downarrow$

(a) Use the Fundamental Theorem of Linear Programming to find a solution; assume a solution exists.

(b) Write down the dual.

(c) Suppose the dual solution is $\boldsymbol{\lambda} = (0, 1)$. If 20 changes to 20.1 and 21 changes to 20.8, what is the corresponding change in the optimal objective value? $\Rightarrow$ change in $z\lambda + 7$

For a sufficiently small relaxation of a primal constraint, the corresponding dual variable measures the per-unit change in the optimal objective value, holding the other constraints fixed.

Given the dual solution $\lambda = (0,1)$, the change in objective value is

$0 \cdot (20.1 - 20) + 1 \cdot (20.8 - 21) = 0 - 0.2 = -0.2$

Tutorial Q3: Knowledge used

Knowledge used

- Fundamental Theorem of Linear Programming: compare objective values at vertices.
- Construction of the dual from a primal in standard form.
- Shadow-price interpretation of dual variables for sufficiently small changes in the right-hand sides of primal constraints.

Tutorial Q3(a): Extreme points and objective values

Solution

The extreme points are

$$(0,0), \qquad (0,3), \qquad (5,0), \qquad \left(\frac{35}{13}, \frac{24}{13}\right).$$

For $f(x,y) = 2x + 7y$,

$$\begin{aligned} f(0,0) &= 0, \\ f(0,3) &= 21, \\ f(5,0) &= 10, \\ f\left(\frac{35}{13}, \frac{24}{13}\right) &= \frac{238}{13}. \end{aligned}$$

Since 21 is the largest,

$$(x^*, y^*) = (0,3)$$

is a solution.

Tutorial Q3(b): Dual problem

Solution

Let $\mu_1, \mu_2 \geq 0$ be the dual variables. The dual is

$$\begin{aligned} \min \quad & 20\mu_1 + 21\mu_2 \\ \text{s.t.} \quad & 4\mu_1 + 3\mu_2 \geq 2, \\ & 5\mu_1 + 7\mu_2 \geq 7, \\ & \mu_1, \mu_2 \geq 0. \end{aligned}$$

Tutorial Q3(c): Change in the optimal value

Solution

A dual variable measures the per-unit change in the optimal primal objective value from a sufficiently small relaxation of the corresponding constraint.

Given

$$\boldsymbol{\lambda} = (\lambda_1, \lambda_2) = (0, 1),$$

the right-hand-side changes are

$$20.1 - 20 = 0.1, \qquad 20.8 - 21 = -0.2.$$

Hence the change in the optimal value is

$$\lambda_1(0.1) + \lambda_2(-0.2) = 0(0.1) + 1(-0.2) = -0.2.$$

Tutorial Q4

Question 4

Write down the dual of each linear programming problem.

(a)

$$\begin{aligned} \max \quad & x_1 + x_2 - 3x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 - 3x_3 \leq 4, \\ & 4x_1 + 5x_2 - 9x_3 \leq 13. \end{aligned}$$

(b) Same objective, but with equality constraints:

$$x_1 + 2x_2 - 3x_3 = 4, \qquad 4x_1 + 5x_2 - 9x_3 = 13.$$

Unrestricted variables

If a primal variable is unrestricted in sign, the corresponding dual constraint holds with equality. Therefore,

$$\max\{\boldsymbol{c} \cdot \boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$$

has dual

min $y \cdot b$ s.t. $yA \geq c, y \geq 0$

$$\min \ \boldsymbol{y} \cdot \boldsymbol{b} \quad \text{s.t.} \quad \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c},$$

If the primal has both $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and unrestricted $\boldsymbol{x}$, then the dual has unrestricted $\boldsymbol{y}$ and $\boldsymbol{y}\boldsymbol{A} = \boldsymbol{c}$.

(a)

$$\begin{aligned} \max \quad & x_1 + x_2 - 3x_3 \\ \text{s.t.} \quad & x_1 + 2x_2 - 3x_3 \le 4, \\ & 4x_1 + 5x_2 - 9x_3 \le 13. \end{aligned}$$

$C = (1, 1, -3)$, $A = \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & -9 \end{pmatrix}$, $b = \begin{pmatrix} 4 \\ 13 \end{pmatrix}$

Let $y = (y_1, y_2)$ is the dual variable vector

The dual: min $y \cdot b$ s.t $yA = c$, $y \ge 0$ because the primal variables are unrestricted in sign.

$y \cdot b = (y_1, y_2) \cdot \begin{pmatrix} 4 \\ 13 \end{pmatrix} = 4y_1 + 13y_2$

$y \cdot A = (y_1, y_2) \cdot \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & -9 \end{pmatrix} = (y_1 + 4y_2, 2y_1 + 5y_2, -3y_1 - 9y_2) = (1, 1, -3)$

$y = (y_1, y_2) \ge 0$

The dual: min $4y_1 + 13y_2$

s.t $y_1 + 4y_2 = 1$

$2y_1 + 5y_2 = 1$

$-3y_1 - 9y_2 = -3$

$y_1, y_2 \ge 0$

(b) With equality constraints, the dual reduces to

min $4y_1 + 13y_2$

s.t $y_1 + 4y_2 = 1$

$2y_1 + 5y_2 = 1$

$-3y_1 - 9y_2 = -3$

Recap: Writing down the dual

Equality constraints

If a primal constraint is an equality, the corresponding dual variable is unrestricted in sign.
In particular,

$$\max\{\boldsymbol{c}\cdot\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}\}$$

has dual

$$\min\ \boldsymbol{y}\cdot\boldsymbol{b} \qquad \text{s.t.} \qquad \boldsymbol{y}\boldsymbol{A} \geq \boldsymbol{c},$$

where $\boldsymbol{y}$ is unrestricted in sign.

Unrestricted variables

If a primal variable is unrestricted in sign, the corresponding dual constraint holds with equality.
Therefore,

$$\max\{\boldsymbol{c}\cdot\boldsymbol{x} : \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b}\}$$

has dual

$$\min\ \boldsymbol{y}\cdot\boldsymbol{b} \qquad \text{s.t.} \qquad \boldsymbol{y}\boldsymbol{A} = \boldsymbol{c},\quad \boldsymbol{y} \geq \boldsymbol{0}.$$

If the primal has both $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ and unrestricted $\boldsymbol{x}$, then the dual has unrestricted $\boldsymbol{y}$ and $\boldsymbol{y}\boldsymbol{A} = \boldsymbol{c}$.

Tutorial Q4: Knowledge used

Knowledge used

The primal variables x_1, x_2, x_3 have no non-negativity restrictions, so the corresponding dual constraints must be equalities.

- If the primal constraints are $\leq$, the dual variables are non-negative.
- If the primal constraints are equalities, the dual variables are unrestricted in sign.

Thus, for unrestricted $\boldsymbol{x}$,

$$\boldsymbol{Ax} \leq \boldsymbol{b} \Rightarrow \boldsymbol{yA} = \boldsymbol{c},\ \boldsymbol{y} \geq \boldsymbol{0},$$

while

$$\boldsymbol{Ax} = \boldsymbol{b} \Rightarrow \boldsymbol{yA} = \boldsymbol{c},\ \boldsymbol{y} \text{ unrestricted}.$$

Tutorial Q4: Matrix notation

Solution

Let

$$\boldsymbol{c} = \begin{pmatrix} 1 & 1 & -3 \end{pmatrix}, \qquad \boldsymbol{A} = \begin{pmatrix} 1 & 2 & -3 \\ 4 & 5 & -9 \end{pmatrix}, \qquad \boldsymbol{b} = \begin{pmatrix} 4 \\ 13 \end{pmatrix}.$$

Part (a) is

$$\max \ \boldsymbol{c} \cdot \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} \leq \boldsymbol{b},$$

and part (b) is

$$\max \ \boldsymbol{c} \cdot \boldsymbol{x} \qquad \text{s.t.} \qquad \boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}.$$

In both parts, $\boldsymbol{x}$ is unrestricted in sign.

Tutorial Q4(a): Dual

Solution

Because the primal variables are unrestricted, the dual constraints are equalities. Because the primal constraints are $\leq$, the dual variables are non-negative.
Therefore, the dual is

$$\begin{aligned} \min \quad & 4y_1 + 13y_2 \\ \text{s.t.} \quad & y_1 + 4y_2 = 1, \\ & 2y_1 + 5y_2 = 1, \\ & -3y_1 - 9y_2 = -3, \\ & y_1, y_2 \geq 0. \end{aligned}$$

Tutorial Q4(b): Dual

Solution

Again, the primal variables are unrestricted, so the dual constraints are equalities. Now the primal constraints are also equalities, so the dual variables are unrestricted in sign.
Hence the dual is

$$\begin{aligned} \min \quad & 4y_1 + 13y_2 \\ \text{s.t.} \quad & y_1 + 4y_2 = 1, \\ & 2y_1 + 5y_2 = 1, \\ & -3y_1 - 9y_2 = -3, \end{aligned}$$

where

y_1, y_2 are unrestricted in sign.